

1. Administrative & Study Identification

Full Study Title: A Modular Phase I/IIa, Open-label, Multi-centre Study to Assess the Safety, Tolerability, Pharmacokinetics, and Preliminary Efficacy of AZD0516 as Monotherapy and in Combination in Participants With Metastatic Prostate Cancer **Short Title/ Acronym:** SEACLIFF **Protocol Number:** D9520C00001 **NCT Number:** NCT07181161 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD0516 (STEAP2 Antibody-Drug Conjugate) and AZD9574 **Phase of Development:** Phase I/IIa **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety and tolerability of AZD0516 as monotherapy and/or in combination with other anti-cancer agents (AZD9574) for the treatment of metastatic prostate cancer. **Secondary Objectives:** To evaluate the preliminary efficacy (e.g., Overall Response Rate, PSA response, Progression-Free Survival), pharmacokinetics (PK), pharmacodynamics, and immunogenicity of AZD0516.

2.2 Background & Rationale To address the clinical need for novel therapies in patients with metastatic castration-resistant prostate cancer (mCRPC) who have progressed despite ongoing androgen deprivation therapy (ADT). AZD0516 is a novel STEAP2-targeting modality (Antibody-Drug Conjugate) designed to target cancer cells that continue to grow despite lowered testosterone levels.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Open-label modular design) **Blinding:** Open-label **Randomization:** N/A (Dose Escalation and Dose Optimisation cohorts)

3.2 Planned Enrollment & Duration **Target N:** 177 participants **Number of Sites:** Multi-centre **Estimated Study Start:** Q4 2025 **Estimated Primary Completion:** >2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, aged 18 years and older, with documented current evidence of metastatic castration-

resistant prostate cancer (mCRPC) with progression at screening. 2. Surgically or medically castrated with serum testosterone levels ≤ 50 ng/dL (≤ 1.75 nmol/L) and ongoing androgen deprivation therapy (ADT) with a GnRH modulator. 3. Measurable PSA ≥ 1 μ g/L (≥ 1 ng/mL) and an ECOG performance status of 0 or 1.

4.2 Exclusion Criteria 1. Previous treatment with a STEAP2 targeting modality, chemotherapeutic agent that inhibits topoisomerase activity, or metabolic enzymes. 2. Cardiovascular disorder (e.g., arrhythmia, uncontrolled hypertension, symptomatic heart failure) or history of non-infectious interstitial lung disease (ILD)/pneumonitis. 3. Active infection (including HBV, HCV, HIV, or Tuberculosis) or Myelodysplastic syndrome (MDS)/Acute Myeloid Leukemia (AML).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Module 1 evaluates AZD0516 monotherapy (Part A: Dose Escalation, Part B: Dose Optimisation, Part C: Efficacy Expansion). Module 2 evaluates AZD0516 in combination with AZD9574 (Part A: Dose Escalation, Part B: Dose Optimisation). **Route of Administration:** Intravenous (AZD0516) and oral/intravenous combination therapies. **Duration of Treatment:** Until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Ongoing androgen deprivation therapy (ADT) with a gonadotropin-releasing hormone (GnRH) modulator is required. **Prohibited:** Concurrent anti-cancer therapies not specified in the protocol, and live attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, PSA, Tumour sample (FFPE), Testosterone check
Baseline	Day 0	Enrollment, First Dose, Safety Labs	
Interim	Cycle-based	Safety Labs, PK/PD blood draws, CT/MRI Efficacy Assessments, Dose Escalation Review	End of Study
Variable	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and severity of Adverse Events (AEs), Serious Adverse Events (SAEs), and Dose-Limiting Toxicities (DLTs) to evaluate safety and tolerability. **Measurement Method:** Clinical assessment graded per CTCAE criteria.

7.2 Secondary & Exploratory Endpoints * Preliminary efficacy including Overall Response Rate (ORR), Progression-Free Survival (PFS), Overall Survival (OS), and PSA response. * Pharmacokinetic (PK) parameters and immunogenicity (Anti-Drug Antibodies).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical, laboratory, and cardiovascular evaluations. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal/External Safety Review Committee (SRC) established for dose escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all treated participants), Efficacy Analysis Set. **Sample Size Justification:** $N=177$ is based on the requirements for a Phase I/IIa modular dose-escalation and expansion design to establish the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). **Handling of Missing Data:** Missing safety data will generally not be imputed; time-to-event efficacy endpoints will use standard censoring rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol compliance and safety. **Audit Trail:** Detailed tracking via an Electronic Data Capture (EDC) system with full audit trails, eCRF locking, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: D9520C00001 **Registry Number:** NCT07181161 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** SEACLIFF **Official Regulatory Title:** A Modular Phase I/IIa, Open-label, Multi-centre Study to Assess the Safety, Tolerability, Pharmacokinetics, and Preliminary Efficacy of AZD0516 as Monotherapy and in Combination in Participants With Metastatic Prostate Cancer

12. Clinical Framework (Metastatic Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Measurable PSA $\geq 1\text{ }\mu\text{g/L}$ and castrate testosterone levels $\leq 50\text{ ng/dL}$ **Medical Methodology:** Targeted STEAP2 Antibody-Drug Conjugate (ADC) Therapy **Optimization Target:** Safety assessment, dose escalation, and optimization of AZD0516

13. Study Procedures & Methodology

Management Pathway: Advanced Prostate Cancer Clinical Pathway **Education Protocol (HCP):** Protocol training and ADC-specific safety management guidelines **Education Protocol (Patient):** Informed consent, treatment schedule counseling, and AE reporting guidelines **Quality Audit Mechanism:** Source data verification and central medical data review for dose-limiting toxicities

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, death, or study withdrawal **Onsite Visit Cadence:** Cycle-based visits corresponding to modular dosing schedules **Remote Monitoring Frequency:** Periodic follow-ups for survival and subsequent treatments **Checklist Review Interval:** Continuous safety monitoring every treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					